

Trigonometric Formula

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June 2026

1 Double Angle Formulas

$$1.1 \quad \sin 2A = 2 \sin A \cos A$$

$$1.2 \quad \cos 2A = \cos^2 A - \sin^2 A$$

$$1.2.1 \quad \cos 2A = 1 - 2 \sin^2 A \implies 2 \sin^2 A = 1 - \cos 2A$$

$$1.2.2 \quad \cos 2A = 2 \cos^2 A - 1 \implies 2 \cos^2 A = \cos 2A + 1$$

2 Triple Angle Formulas

$$2.1 \quad \sin 3A = 3 \sin A - 4 \sin^3 A$$

$$2.2 \quad \cos 3A = 4 \sin^3 A - 3 \cos A$$

$$2.3 \quad \tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

$$2.4 \quad \cot A = \frac{3 \cot A - \cot^3 A}{1 - 3 \cot^2 A}$$

3 Expression $\sin$, $\cos$, $\tan$, $\cot \rightarrow \tan$

$$3.1 \quad \sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$$

$$3.2 \quad \cos A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$3.3 \quad \tan A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$3.4 \quad \cot A = \frac{\cot^2 A - 1}{2 \cot^2 A}$$